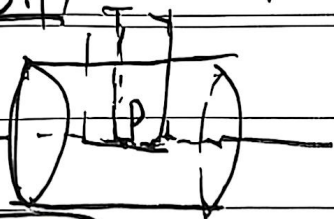


13.1

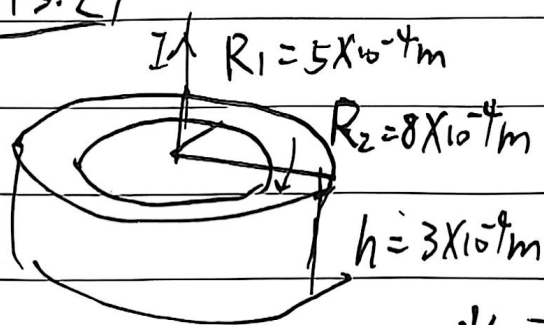
$$n = 1000$$



$$B \cdot l = \mu_0 I \cdot n l$$

$$\Rightarrow I = \frac{B}{\mu_0 n} = 0.33 \text{ A}$$

13.2



$$R_1 = 5 \times 10^{-4} \text{ m}$$

$$R_2 = 8 \times 10^{-4} \text{ m}$$

$$h = 3 \times 10^{-4} \text{ m}$$

$$(1) \quad \vec{M} = 0 \quad \vec{M} = \chi_m \vec{H} \quad \vec{H} = \frac{B}{\mu_0}$$

H_c 时, 开始转动

$$H_c \cdot 2\pi r = I$$

当 $I = H_c \cdot 2\pi R_1$ 时, 开始转动

$$I = 0.5 \text{ A}$$

$$(2) \quad I_{\max} = H_c \cdot 2\pi R_2 = 0.8 \text{ A}$$

13.4 $N = 300$ $2\pi r = 0.3 \text{ m}$ $S_{AF} = 1 \times 10^{-4} \text{ m}^2$

$$I = 0.032 \text{ A} \quad \Phi = 2 \times 10^{-6} \text{ Wb}$$

$$(1) \quad \Phi = B \cdot S \Rightarrow B = 0.02 \text{ T}$$

$$\odot H \cdot 2\pi r = NI \Rightarrow H = 32 \text{ A/m}$$

$$(2) \quad \vec{j}_m = \vec{M} \times \vec{n}$$

$$\vec{M} = \chi_m \vec{H} \Rightarrow \vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} \Rightarrow \vec{M} = \frac{\vec{B}}{\mu_0} - \vec{H} = 15883.5 \text{ A/m}$$

$$\vec{j}_m = 15883.5 \text{ A/m}$$

$$3) \quad B = \mu_0 \mu_r H \Rightarrow \mu_r = \frac{B}{\mu_0 H} = 497.36 \text{ T} \cdot \text{m/A}$$

13.8

$$\rho = 7.8 \times 10^3$$

$$P_m = 1.8 \times 10^{-23} \text{ A} \cdot \text{m}^2 \quad L = 5 \times 10^{-2} \text{ m} \quad S = 1 \times 10^{-4} \text{ m}^2 \quad M = 55.85 \times 10^{-3} \text{ kg}$$

$$(1) P_{\text{总}} = \frac{\rho S L N P_m}{M} = 7.57 \text{ A} \cdot \text{m}^2$$

$$(2) \begin{array}{l} \uparrow B = 1.5 \text{ T} \\ \vec{M} = \vec{P}_{\text{总}} \times \vec{B} \\ |\vec{M}| = 11.4 \text{ N} \cdot \text{m} \end{array}$$

复习题: 12.5 12.19 12.21 12.32 13.3

12.5 题表理解

$$B_1 = \frac{\mu_0 I_1}{2R} \cdot \frac{ab}{2\pi r}$$

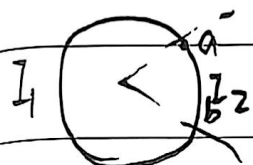
$$B_2 = \frac{\mu_0 I_2}{2R} \cdot \frac{ab}{2\pi r}$$

还可以化简

$$I_1 r_1 = I_2 r_2$$

$$\Rightarrow I_1 ab = I_2 ab$$

考虑方向

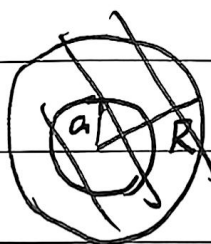


$$\vec{B} = B_1 + B_2 = \frac{\mu_0 I}{2R} \left(\frac{I_1 ab}{2\pi r} + \frac{I_2 ab}{2\pi r} \right) = 0$$

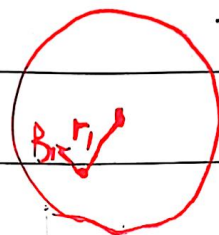
12.19

$$j = \frac{I}{\pi R^2} \quad d\vec{B} = \frac{\mu_0}{4\pi} \frac{j d\vec{L} \times \vec{r}}{r^3}$$

如图取对称的2个电流



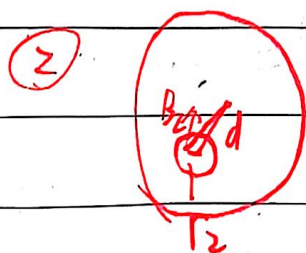
叠加法:



$$d\vec{B}_1 = \frac{\mu_0}{4\pi} \frac{j d\vec{L}_1 \times \vec{r}_1}{r_1^3}$$

$$B_1 \cdot 2\pi r_1 = \mu_0 \frac{I}{R^2} r_1^2$$

$$\Rightarrow \vec{B}_1 = \frac{\mu_0 I}{2\pi(R^2 - a^2)} r_1 (\hat{L} \times \vec{r}_1) = \frac{\mu_0 I}{2\pi(R^2 - a^2)} \hat{L} \times \vec{r}_1$$



$$\vec{B}_2 = -\mu_0 I$$

$$B_2 \cdot 2\pi r_2 = \mu_0 \frac{I a^2}{R^2 a^2} \frac{r_2^2}{a^2}$$

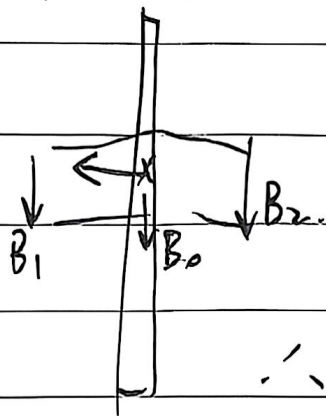
$$\Rightarrow \vec{B}_2 = \frac{\mu_0 I}{2\pi(R^2 - a^2)} r_2 (\hat{L} \times \vec{r}_2) = \frac{\mu_0 I}{2\pi(R^2 - a^2)} \hat{L} \times \vec{r}_2$$

$$\vec{B} = \vec{B}_1 - \vec{B}_2 = \frac{\mu_0 I}{2\pi(R^2 - a^2)} \hat{L} \times (\vec{r}_1 - \vec{r}_2) = \frac{\mu_0 I}{2\pi(R^2 - a^2)} \hat{L} \times \vec{d} = \frac{\mu_0 I d}{2\pi(R^2 - a^2)}$$

12.21

$$\textcircled{1} B_2 \cdot dl - B_1 \cdot dl = \mu_0 j \cdot dl$$

$$\Rightarrow j = \frac{1}{\mu_0} (B_2 - B_1)$$



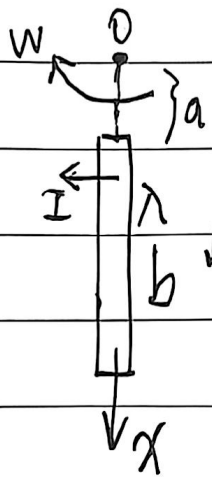
$$\textcircled{2} B_1 = B_0 - B_i \quad B_2 = B_0 + B_i$$

$$\Rightarrow B_0 = \frac{B_1 + B_2}{2}$$

$$\therefore F = B_0 j \cdot 1 \cdot 1 = \frac{1}{2\mu_0} (B_2^2 - B_1^2) \quad \text{水平向左}$$

12.32

$$dq = \lambda dl \quad I = dq \cdot l = \frac{w}{2\pi} \lambda dl$$



$$\therefore dB = \frac{\mu_0 I}{4\pi l} = \frac{\mu_0 I}{4\pi l} = \frac{\mu_0 w \lambda}{4\pi l} dl$$

$$\therefore B = \int_a^{a+b} \frac{\mu_0 I w \lambda}{4\pi l} dl = \frac{\mu_0 w \lambda}{4\pi} \ln \frac{a+b}{a} \quad \text{垂直纸面向内}$$

$$dP = IS = \frac{w}{2\pi} \lambda dl \cdot \lambda l^2 = \frac{w \lambda}{2} l^2 dl$$

$$\therefore P = \int_a^{a+b} dP = \frac{w \lambda}{6} [(a+b)^3 - a^3]$$

13.3

$$2\pi r = 0.61m \quad d = 0.01m \quad N = 1000$$

$$B = 0.18T, \quad B = \mu H$$

$$H(2\pi r - d) = NI \Rightarrow H = \frac{NI}{2\pi r - d}$$

$$\therefore u = \frac{B}{N \cdot I} (2\pi r - d) = 7.2 \times 10^{-5} T \cdot m/A$$

☆: 空气处也有 H, (+B法向连续)

$$H_1(2\pi r - d) + H_2 d = NI$$

$$H_1 = \frac{B}{\mu} \quad H_2 = \frac{B}{\mu_0}$$

$$u = \frac{NI - \frac{B}{\mu_0} d}{2\pi r - d} = \frac{B(2\pi r - d)}{NI - \frac{B}{\mu_0} d} = 1.60 \times 10^{-3} T \cdot m/A$$